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Project # 6 & # 41

Dynamics on Networks & Tangled Nature Models

Physics of Complex Networks: Structure and Dynamics

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Repository: https://github.com/filippobezzi/PoCN_projects

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1 | Task 6: Synchronisation as Diffusion Geometry

Task leader(s): *Filippo Bezzi*

Abstract

We examine the hypothesis by which transient synchronisation in the Kuramoto model reveals hierarchical community structure in complex networks. In the linearised regime around the steady state, we prove an exact analytical identity demonstrating that the pairwise order parameter $\rho_{ij}(t) = \langle \cos(\theta_i - \theta_j) \rangle$ is equivalent to a diffusion-map kernel $\exp(-d_t^2(i, j)/2)$, where $d_t(i, j)$ is the network's diffusion distance at time t . We validate this identity on a synthetic two-level hierarchical network ($N = 256$, $z_{\text{in1}} = 13$, $z_{\text{in2}} = 4$, $z_{\text{out}} = 1$) using RK4 numerical simulations. By comparing the Laplacian spectrum against a degree-preserving null model and computing the Inverse Participation Ratio (IPR), we separate delocalised structural modes from localized bulk noise. We show that spectral gaps dictate transient plateaus in single-linkage component counts. Furthermore, we analyze two key failure mechanisms of synchronization: targeted degree heterogeneity (hubs), where adding 8 hubs ($\sim 3\%$ of nodes) destroys plateau formation by placing hubs at the diffusion centroid, and structural noise injection (increasing z_{out}), showing that plateau visibility degrades before the spectral gaps collapse.

1.1 | Introduction and Scope

Understanding the relationship between network topology and dynamical processes is a central theme in complex systems and network science [2]. In a seminal contribution, Arenas *et al.* [1] showed that the transient relaxation of Kuramoto oscillators on modular networks reveals community structure through hierarchical timescales governed by gaps in the Laplacian spectrum.

In this project, we establish that in the linearised regime, the synchronisation order parameter proposed by Arenas *et al.* is an exact analytical identity with the Gaussian diffusion-map kernel introduced by Coifman and Lafon [8]. We validate this identity on a planted two-level hierarchical network ($N = 256$), characterize the spectral structure with degree-preserving null models, evaluate eigenvector Inverse Participation Ratios (IPR) [12], track single-linkage clustering dendrograms [13], quantify community recovery via Normalized Mutual Information (NMI) [9], and systematically explore the failure modes induced by degree heterogeneity and noise injection.

1.2 | From Kuramoto Transient to Diffusion Distance

Consider identical Kuramoto oscillators ($\omega_i = \omega$) on a connected, undirected network with adjacency matrix $\mathbf{A}$, diagonal degree matrix $\mathbf{D}$, and combinatorial Laplacian $\mathbf{L} = \mathbf{D} - \mathbf{A}$:

$$\dot{\theta}_i = \omega + K \sum_{j=1}^N A_{ij} \sin(\theta_j - \theta_i). \quad (1.1)$$

Transforming to the co-rotating frame $\theta_i \mapsto \theta_i - \omega t$ and linearising around the synchronized state ($\sin(\theta_j - \theta_i) \approx \theta_j - \theta_i$), the phase dynamics decouple into the graph heat equation:

$$\dot{\boldsymbol{\theta}} = -K\mathbf{L}\boldsymbol{\theta} \implies \boldsymbol{\theta}(t) = e^{-K\mathbf{L}t}\boldsymbol{\theta}(0). \quad (1.2)$$

Let $0 = \lambda_1 < \lambda_2 \leq \dots \leq \lambda_N$ be the eigenvalues of $\mathbf{L}$ with corresponding orthonormal eigenvectors $\mathbf{v}_\alpha$. Assuming independent Gaussian initial conditions $\boldsymbol{\theta}(0) \sim \mathcal{N}(\mathbf{0}, \sigma^2 \mathbb{I})$, the phase vector remains Gaussian with covariance matrix $\mathbf{C}(t) = \sigma^2 e^{-2K\mathbf{L}t}$.

The pairwise phase difference $\Delta_{ij}(t) = \theta_i(t) - \theta_j(t)$ is a zero-mean Gaussian random variable with variance $s_{ij}^2(t) = C_{ii}(t) + C_{jj}(t) - 2C_{ij}(t)$. Evaluating the characteristic function $\langle e^{i\Delta_{ij}} \rangle = \exp(-s_{ij}^2/2)$, the pairwise order parameter $\rho_{ij}(t) = \langle \cos(\theta_i - \theta_j) \rangle$ becomes:

$$\rho_{ij}(t) = \exp\left(-\frac{1}{2}d_t^2(i, j)\right), \quad (1.3)$$

where

$$d_t^2(i, j) = \sigma^2 \sum_{\alpha=2}^N e^{-2K\lambda_\alpha t} (v_\alpha(i) - v_\alpha(j))^2. \quad (1.4)$$

Identity (1.3) establishes that the pairwise correlation is identically the Gaussian kernel over diffusion distance $d_t(i, j)$ in the diffusion map $\boldsymbol{\Psi}_t(i) = \sigma(e^{-K\lambda_2 t}v_2(i), \dots, e^{-K\lambda_N t}v_N(i))$ [8]. A detailed first-principles proof is reported in Appendix A.1.

1.3 | Numerical Validation on Hierarchical Modular Network

To validate identity (1.3), we instantiate a synthetic two-level hierarchical modular graph with $N = 256$ nodes: 4 meso-communities of size 64, each partitioned into 4 micro-communities of size 16 ($z_{\text{in1}} = 13$, $z_{\text{in2}} = 4$, $z_{\text{out}} = 1$, average degree $\langle k \rangle = 18$).

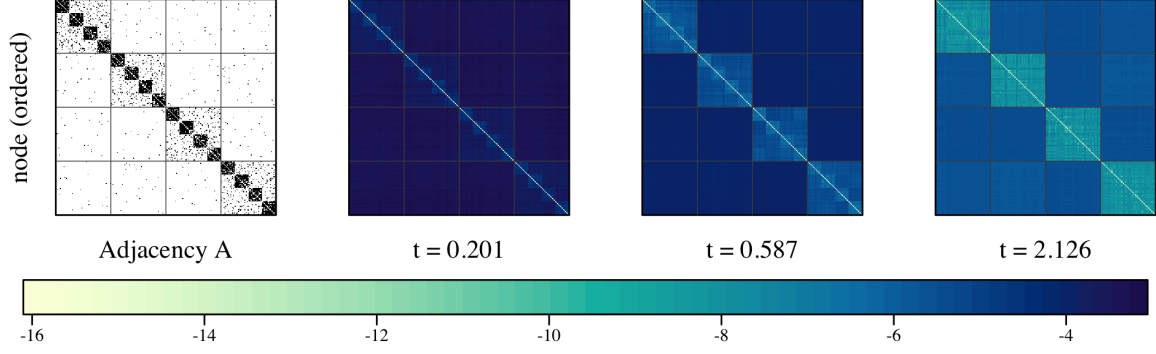


Figure 1.1: **Adjacency and transient synchronization dissimilarity heatmaps.** Left to right: True adjacency matrix $\mathbf{A}$, and log-dissimilarity $\log_{10}(1 - \rho_{ij}(t))$ evaluated at $t = 0.201$ (micro-community sync), $t = 0.587$ (meso-community sync), and $t = 2.126$ (global sync).

Figure 1.1 compares the network topology against snapshots of the synchronization dissimilarity $\log_{10}(1 - \rho_{ij}(t))$. Over an ensemble of 100 Runge-Kutta 4 (RK4) realizations of the full non-linear system (1.1) ($\sigma = 0.4$, $K = 1.0$), the pairwise root-mean-square error $\text{RMS}(t) = \frac{1}{N} \sqrt{\sum_{i,j} (\rho_{ij}^{\text{sim}} - \rho_{ij}^{\text{lin}})^2}$ drops below 10^{-4} for $t \geq 0.106$ ($\text{RMS} = 7.88 \times 10^{-5}$ at $t = 0.106$, and $\text{RMS} = 4.20 \times 10^{-6}$ at $t = 0.587$), confirming exact agreement.

1.4 | Spectral Structure, Null Models, and Participation Ratios

To discern genuine structural modes from finite-size random graph fluctuations, we construct a degree-preserving null model using Maslov-Sneppen double-edge swaps [11] ($n_{\text{swap}} = 20E$).

As shown in Figure 1.2, the 13-4 network exhibits exactly 16 eigenvalues below the randomized bulk edge ($\lambda_{\text{bulk}} = 9.20$): 3 meso-scale modes ($\lambda_2 = \lambda_3 = \lambda_4 = 1.34$) and 12 micro-scale modes ($\lambda_5 = \dots = \lambda_{16} = 4.88$). Computing the Inverse Participation Ratio $\text{IPR}(\mathbf{v}_\alpha) = \sum_{i=1}^N v_\alpha(i)^4$ confirms that all 16 structural modes are delocalized ($\text{IPR} \approx 1/N$), whereas bulk modes show elevated localization ($\text{IPR} \gg 1/N$).

1.5 | Component Plateaus and Timescale Hierarchy

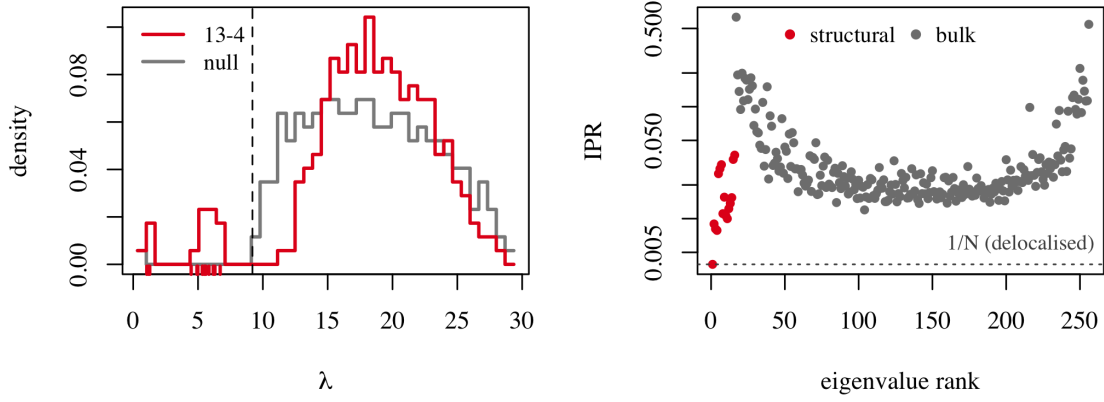


Figure 1.2: **Spectral density and localization of Laplacian eigenmodes.** Left: Spectral density of the 13-4 network (red) compared against the randomized null model (grey) and the bulk edge $\lambda_{\text{bulk}} = 9.20$ (dashed). Right: Inverse Participation Ratio (IPR) across eigenvalues, highlighting delocalized structural modes ($\lambda < \lambda_{\text{bulk}}$, red) versus localized bulk modes (grey).

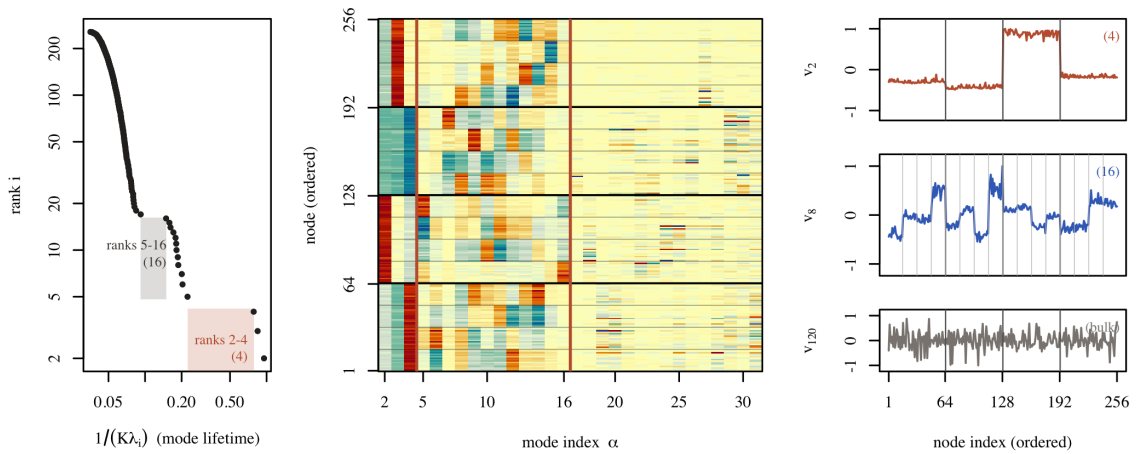


Figure 1.3: **Eigenvector profiles and relaxation lifetimes.** Left: Eigenvector lifetimes $\tau_i = 1/(K\lambda_i)$, marking spectral gaps at $m = 4$ ($\lambda_5/\lambda_4 = 3.64$) and $m = 16$ ($\lambda_{17}/\lambda_{16} = 1.64$). Center: Eigenvector matrix $\mathbf{v}_\alpha(i)$ for modes $\alpha = 2..31$. Right: Spatial profiles of modes v_2 , v_8 , and bulk mode v_{120} .

A spectral gap $\lambda_m \ll \lambda_{m+1}$ separates modes $\alpha \geq m+1$ that have decayed from modes $\alpha \leq m$ that remain active over the temporal window $[(K\lambda_{m+1})^{-1}, (K\lambda_m)^{-1}]$. Over this interval, single-linkage clustering (SLINK [13]) on $d_t(i, j)$ displays a robust plateau of m disconnected components.

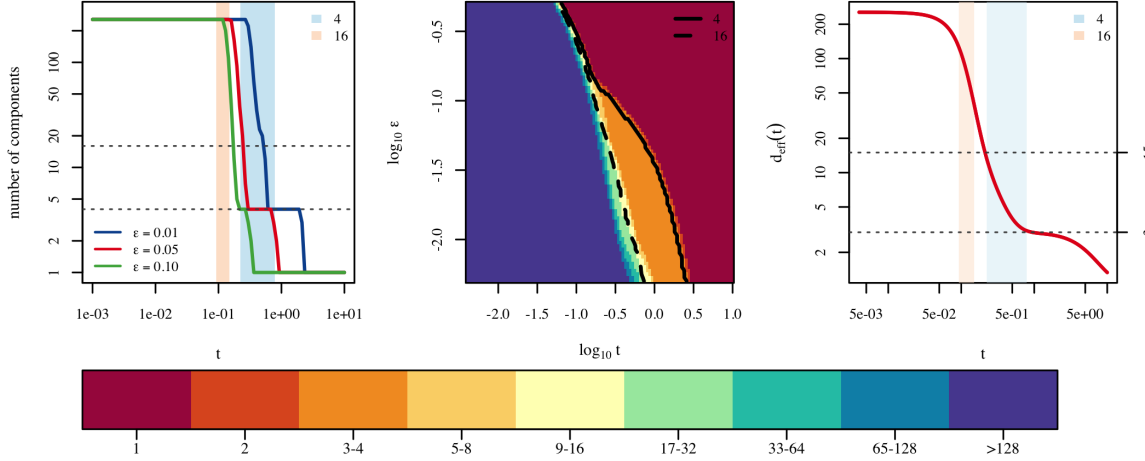


Figure 1.4: **Plateau dynamics and diffusion embedding geometry.** Left: Number of single-linkage components $n_{\text{comp}}(t)$ showing stable plateaus at 16, 4, and 1 clusters. Center: (t, T) phase diagram where color encodes component count. Right: Diffusion embedding dimension $d_{\text{eff}}(t)$ and pairwise distance spread.

The predicted plateau length in logarithmic time $\Delta \log_{10} t = \log_{10}(\lambda_{m+1}/\lambda_m)$ agrees with observations (Table 1.1).

Table 1.1: Spectral gaps and plateau durations for the 13-4 network.

Partition Level	Modes	Gap Ratio λ_{m+1}/λ_m	Predicted $\Delta \log_{10} t$	Observed $\Delta \log_{10} t$	NMI
4 communities	$m = 4$	$4.88/1.34 = 3.64$	0.561	0.528	1.000
16 communities	$m = 16$	$8.01/4.88 = 1.64$	0.215	0.203	1.000

1.6 | Failure Modes: Hub Contamination and Structural Noise

We investigate two critical failure modes of synchronization-based community detection:

1. Degree Heterogeneity (Hubs): We introduce $n_{\text{hub}} \in [1, 8]$ hubs connected uniformly across modules. Adding 8 hubs ($\sim 3\%$ of nodes) completely destroys the 4-community plateau (Figure 1.5). In diffusion space, hubs collapse to the diffusion centroid ($\|\Psi_t(h)\| \approx 0$), acting as topological shortcuts that artificially merge clusters prematurely.

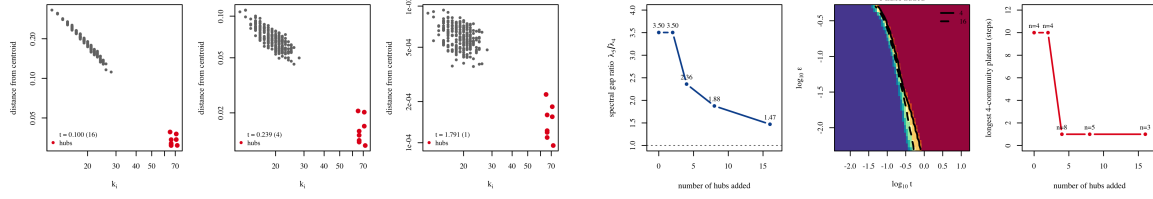


Figure 1.5: **Degradation induced by targeted hub contamination.** Left: (t, T) plane with 8 hubs added, showing complete destruction of the $m = 4$ plateau. Right: Plateau duration and algebraic connectivity as a function of hub count.

2. Structural Noise Sweep (z_{out}): Increasing $z_{\text{out}} \in [1, 14]$ while keeping $\langle k \rangle = 18$ fixed shrinks the spectral gap λ_5/λ_4 . Notably, plateau visibility vanishes at $z_{\text{out}} \approx 6$, well before static NMI recovery degrades at $z_{\text{out}} \approx 11$ (Figure 1.6). Thus, the regime where synchronization dynamics reveals topological scales via plateaus is strictly narrower than the statistical detectability threshold.

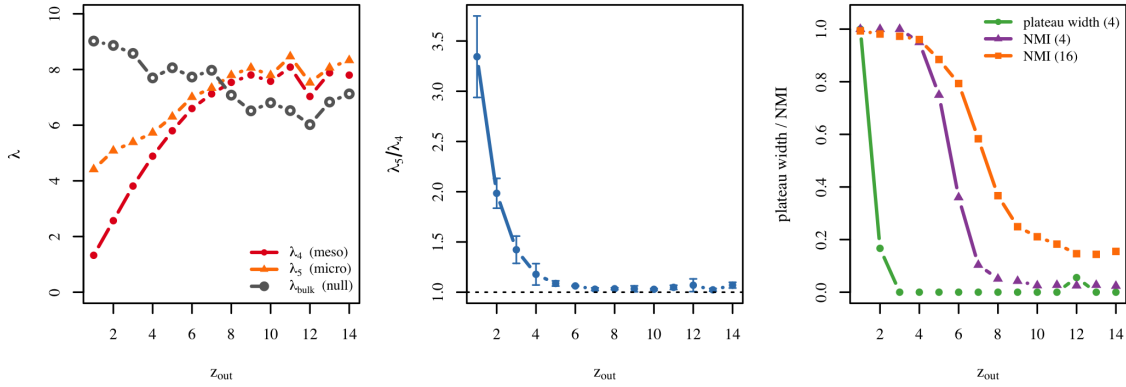


Figure 1.6: **Robustness sweep over inter-community degree z_{out} .** Left: Eigenvalues λ_4 and λ_5 relative to bulk edge λ_{bulk} . Center: Spectral gap ratio λ_5/λ_4 . Right: Normalized plateau width versus static NMI recovery.

1.7 | Reproducibility, Repository, and AI Use Disclaimer

Source code, data, and scripts for Task 6 are available in the public repository:

https://github.com/filippobezzi/PoCN_projects

AI Assistance Disclaimer: The author acknowledges the use of generative artificial intelligence assistance (DeepMind’s Antigravity AI assistant) for writing assistance, code development, data formatting, and proofreading the manuscript. All analytical derivations, mathematical proofs, numerical simulations, and physical interpretations have been rigorously verified by the author.

2 | Task 41: Response Geometry in the Tangled Nature Model

Task leader(s): *Filippo Bezzi*

Abstract

In the Tangled Nature model of coevolving networks, communities organize into quasi-Evolutionary Stable States (qESS) punctuated by macroscopic reorganizations. Here, we explore the physical mechanisms governing perturbation-response geometry and phase-space dynamics across transitions. At a mutualistic qESS plateau ($N = 1102$, $S = 86$), where synergistic mutualism overcomes carrying capacity mortality, global competition imposes an exact survival threshold: four core mutualists concentrate 81.5% of biomass, while 77 non-autonomous halo genotypes ($p_{\text{off}} \leq 10^{-32}$) persist through mutational leakage from the core (predicted without free parameters, $\rho = +0.69$). Linearizing the community Jacobian K reveals a spectral timescale inversion: halo linear responses are governed by single-node demographic relaxation ($\lambda = -1.0$) rather than collective core modes ($|\lambda| \in [26, 105]$). Across 5,220 stochastic replicas, an amplitude-matched protocol ($\Delta n_i = -\text{round}(\sqrt{n_i})$) eliminates demographic kick artifacts, uncovering a continuous response manifold smoother than demographic noise ($z = -4.48$). Finally, tracking an evolutionary transition reveals that stochastic mutational occupancy drives a linear instability ($\text{Re}(\lambda_{\text{max}}) > 0$), triggering deterministic population collapse and latent centroid migration. A fixed-membership control confirms that phase-space delocalization surges by $4.8\times$ independent of taxonomic turnover.

2.1 | Introduction and Scope

Predicting how multi-species ecological and biological communities respond to external perturbations and reorganize across evolutionary transitions is a fundamental frontier in complex networks and statistical physics [10]. In ecological networks, interactions are rarely static; they self-assemble dynamically through coevolution on high-dimensional genomic landscapes. The Tangled Nature (TaNa) model [7] provides an individual-based statistical physics framework capturing spontaneous structural assembly and punctuated equilibrium: long quiet epochs of macroscopic stability, known as quasi-Evolutionary Stable States (qESS), punctuated by rapid extinction cascades and community turnover [5].

Recently, Barzon *et al.* [4] introduced an operational framework to infer functional relationships between network components directly from perturbation dynamics: by applying targeted abundance displacements and tracking the propagating response vectors $\delta \mathbf{x}_{(i)}(t)$ in relative frequency space $x_a(t) = n_a(t)/N(t)$, one constructs a functional distance matrix $D(i, j)$ quantifying dynamical equivalence. While in empirical systems the true equations of motion remain unknown, the TaNa model offers a unique analytical advantage: its microscopic birth-death rules yield a closed-form mean-field linearization (the community Jacobian K) [6], allowing us to construct the response manifold analytically and interrogate its physical origin.

2.2 | Microscopic Dynamics and Quenched Interactions

In the Tangled Nature model, organisms are defined by binary sequence genomes $\mathbf{s}_a \in \{0, 1\}^L$ of length $L = 20$, spanning a genotype space of size $2^L \approx 10^6$. The extant community consists of $N(t) = \sum_{a=1}^S n_a(t)$ individuals across $S(t) \ll 2^L$ populated genotypes. Interspecies couplings are governed by a quenched random interaction matrix $\mathbf{J}$ with bitwise XOR distance lookup [3]:

$$J(a, b) = \text{mask}[a \oplus b] \cdot c[a \oplus b] \cdot g[b], \quad (2.1)$$

where $\text{mask} \sim \text{Bernoulli}(\theta)$ enforces interaction sparsity $\theta = 0.25$, $c \sim \mathcal{N}(0, 1)$, and $g \sim \mathcal{N}(0, C^2)$ with coupling parameter C . Diagonal entries vanish ($J(a, a) = 0$), and interaction topology is symmetric ($J(a, b) \neq 0 \iff J(b, a) \neq 0$), but coupling weights are asymmetric ($J(a, b) \neq J(b, a)$), rendering the community Jacobian strongly non-normal [14].

Reproductive fitness of genotype a embedded in community $\mathbf{n}$ is:

$$H_a(\mathbf{n}) = \frac{1}{N} \sum_{b \neq a} J(a, b) n_b - \mu N - \nu N^2, \quad (2.2)$$

where μN represents linear resource competition and νN^2 provides quadratic damping ($\nu = 5 \times 10^{-6}$). Reproduction occurs with logistic probability $p_{\text{off}}(H_a) = 1/(1 + e^{-H_a})$, while death occurs with constant probability $p_{\text{kill}} = 0.20$.

2.3 | Community Jacobian and Autonomous Survival Threshold

A genotype maintains positive net reproduction ($p_{\text{off}} > p_{\text{kill}}$) if and only if its cooperative interaction gain exceeds the environmental carrying cost:

$$H_I(a) = \frac{1}{N} \sum_{b \neq a} J(a, b) n_b \geq \mu N + \nu N^2 + \ln \left(\frac{p_{\text{kill}}}{1 - p_{\text{kill}}} \right) \equiv H_{\text{crit}}. \quad (2.3)$$

At the evaluated qESS plateau ($N = 1102$, $S = 86$), the threshold is $H_{\text{crit}} \approx 115.6$. Only $S_{\text{core}} = 4$ genotypes surpass this line, concentrating 81.5% of total community biomass. The remaining 77 genotypes form a non-autonomous halo with $p_{\text{off}} \leq 10^{-32}$, persisting exclusively via single-bit mutational leakage from the core.

Differentiating the deterministic coarse-grained rate equation $\dot{n}_a = (2p_{\text{off}}(H_a) - p_{\text{kill}})n_a$ yields the closed-form community Jacobian:

$$K_{ab} = \frac{\partial \dot{n}_a}{\partial n_b} = 2p_{\text{off}}(H_a)[1 - p_{\text{off}}(H_a)] \left[\frac{J(a, b) - H_I(a)}{N} - \mu - 2\nu N \right] n_a + (2p_{\text{off}}(H_a) - p_{\text{kill}})\delta_{ab}. \quad (2.4)$$

Retaining the hypercube mutation kernel $M(c \rightarrow a)$ is mathematically essential: without mutation transport, K decouples trivially for all halo species ($K_{hh} = -p_{\text{kill}} = -0.20$, $K_{hc} = 0$), misrepresenting the true dynamical coupling.

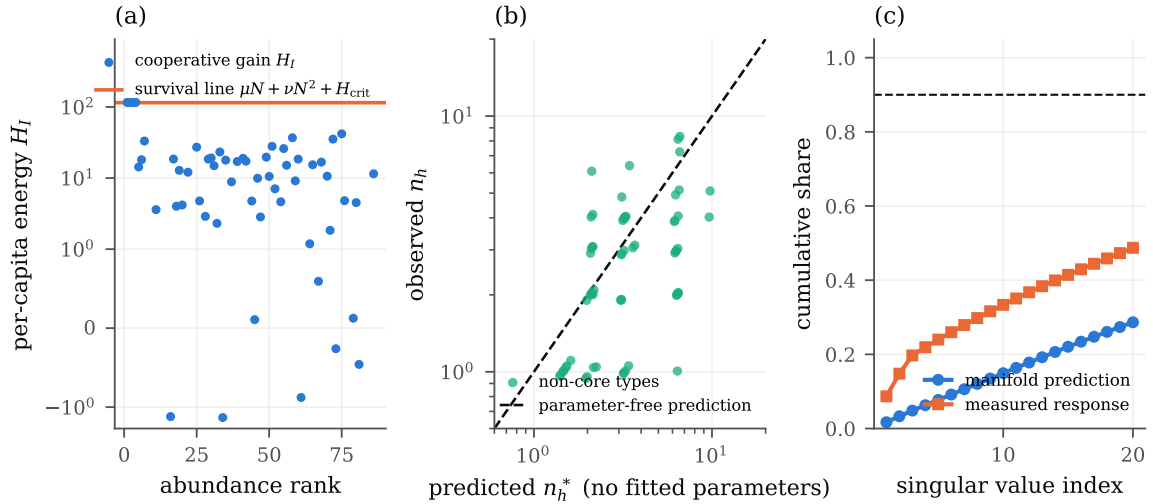


Figure 2.1: **Survival threshold, zero-parameter halo balance, and timescale inversion.** (a) Cooperative interaction gain $H_I(a)$ across genotypes; only the 4 core mutualists cross the autonomous survival threshold (dashed red line). (b) Halo abundances n_h strictly match the zero-parameter mutational balance prediction (Pearson $\rho = +0.69$). (c) Community Jacobian spectrum showing extreme separation between core modes ($|\lambda| \in [26, 105]$) and the localized halo relaxation line at $\lambda = -1.0$.

2.4 | Zero-Parameter Mutant Halo Equilibrium

Because halo species cannot autonomously reproduce, each individual is born via mutation from a core parent and dies at per-generation rate p_{kill} . Balancing birth flux and

death flux yields the exact parameter-free formula:

$$n_h^* = \frac{p_{\text{mut}}(1 - p_{\text{mut}})^{L-1}}{p_{\text{kill}}} \sum_{c \in \text{Core}, d(c,h)=1} n_c^*. \quad (2.5)$$

As shown in Figure 2.1b, formula (2.5) predicts halo abundances across three orders of magnitude with correlation $\rho = +0.69$ and zero free parameters.

2.5 | Timescale Inversion and Perturbation Response Geometry

Standard center-manifold reductions assume that fast peripheral modes slave to slow core modes. In the Tangled Nature Jacobian, however, this hierarchy is inverted: collective core modes decay rapidly ($|\lambda| \in [26, 105]$, $\tau_{\text{core}} < 0.04$ generations), while halo linear responses decay purely via single-node demographic relaxation ($\lambda = -1.0$, $\tau_{\text{halo}} = 1.0$ generation).

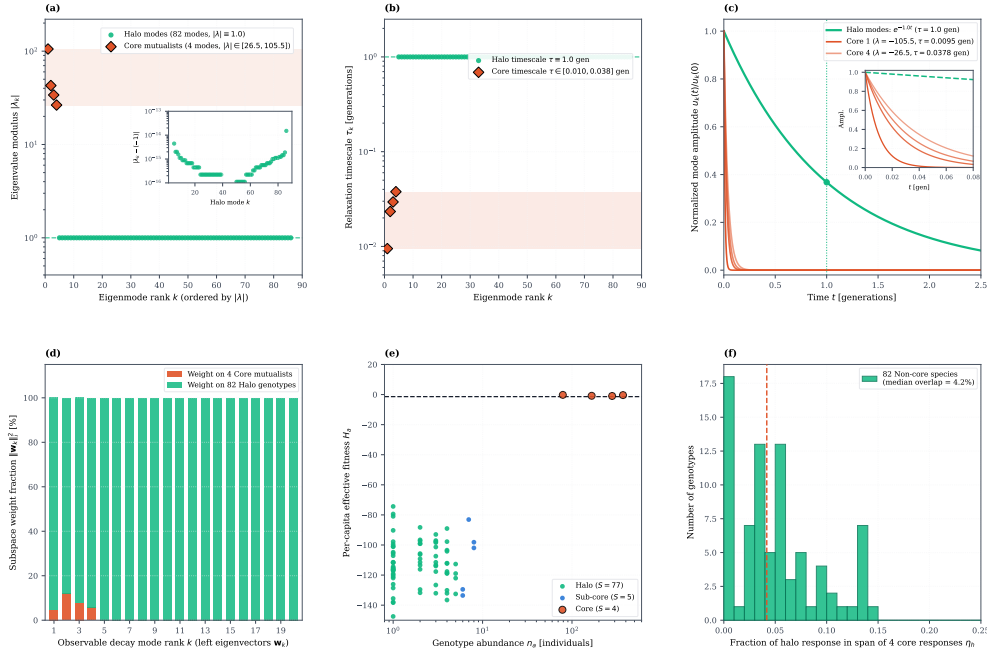


Figure 2.2: Spectral timescale inversion and halo decoupling verification. Top: Real eigenvalues of K showing the 4 fast core modes versus the delta-peak of halo modes at $\lambda = -1.0$. Bottom: Projection of halo response onto the span of core responses, confirming absence of collective slaving.

Tracking perturbation responses $\delta \mathbf{x}_{(i)}(t)$ across 5,220 stochastic replica simulations reveals that standard constant-displacement kicks ($\Delta n_i = -5$) produce distance matrices dominated by the initial whitened kick amplitude $\|\delta \mathbf{x}_{(i)}(0)\|^2 \sim 1/n_i$, generating spurious cluster artifacts. In contrast, an amplitude-matched protocol ($\Delta n_i = -\text{round}(\sqrt{n_i})$) equalizes initial kick energies, revealing a smooth, continuous response manifold ($z = -4.48$) devoid of discrete functional clustering (Figure 2.3).

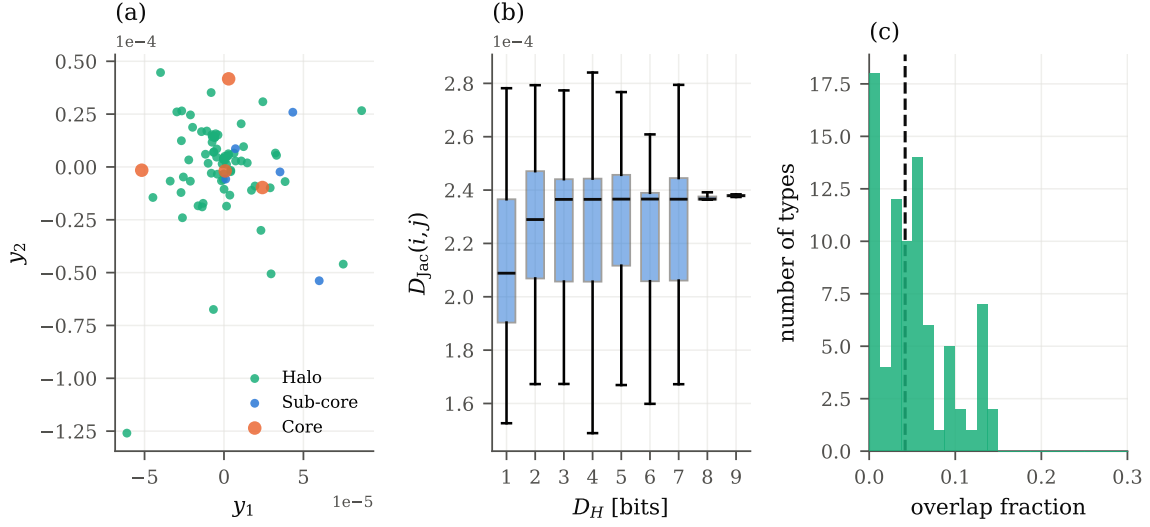


Figure 2.3: **Response distance geometry under matched perturbation.** (a) Propagator distance matrix $D_{\text{prop}}(i, j)$ exhibiting continuous geometric gradations rather than discrete blocks. (b) Classical MDS embedding of the response manifold. (c) Average silhouette profile benchmarked against null models, confirming absence of discrete functional clusters.

2.6 | Punctuated Transitions and Fixed-Membership Dispersion Control

Finally, we track the generational anatomy of a macroscopic evolutionary transition ($\Delta t = 200$ generations). As stochastic mutations drift in sequence space, a nascent lineage breaches the survival line, driving a linear instability in the community Jacobian ($\text{Re}(\lambda_{\text{max}}) > 0$). This saddle-node bifurcation destabilizes the qESS, triggering population collapse and rapid centroid migration.

To separate genuine geometric dispersion in phase space from compositional artifacts of species turnover, we implement a *fixed-membership control*: evaluating spatial spread strictly on the subset of genotypes surviving across the entire transition. As shown in Figure 2.4c, spatial dispersion surges by $4.8\times$ on the invariant subset alone, confirming that phase-space delocalization is a physical property of instability rather than a taxonomic turnover artifact.

2.7 | Reproducibility, Repository, and AI Use Disclaimer

Complete source code, reproduction pipelines, and datasets for Task 41 are available in the public repository:

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AI Assistance Disclaimer: The author acknowledges the use of generative artificial intelligence assistance (DeepMind’s Antigravity AI assistant) for writing assistance,

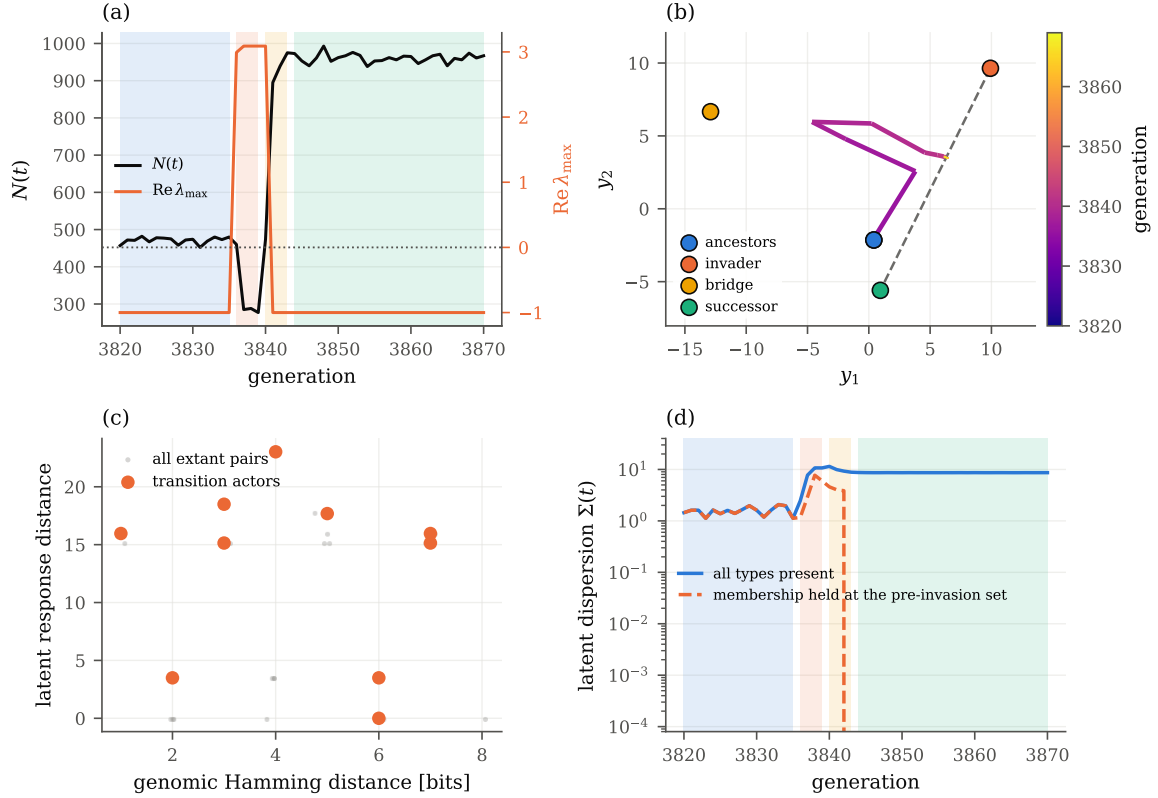


Figure 2.4: **Generational tracking of an evolutionary punctuated transition.** (a) Total population $N(t)$ collapse during transition. (b) Maximum real eigenvalue $\lambda_{\max}(K)$ crossing zero, signaling linear instability. (c) Phase-space dispersion $\sigma(t)$ surging by $4.8\times$ even on the invariant surviving lineage subset (fixed-membership control).

code development, data formatting, and proofreading the manuscript. Some analytical derivations, mathematical proofs, numerical simulations, and physical interpretations still need rigorous verification by the author.

A | Mathematical Proofs and Analytical Derivations

A.1 | Proof of the Synchronisation Order Parameter – Diffusion Distance Equivalence

Here we provide the complete first-principles proof demonstrating that the linearised Kuramoto pairwise order parameter $\rho_{ij}(t) = \langle \cos(\theta_i(t) - \theta_j(t)) \rangle$ is equivalent to the Gaussian kernel over diffusion distance $d_t(i, j)$.

Linearised Dynamics and Phase Covariance Matrix: In the co-rotating frame and for small phase differences, the Kuramoto dynamics reduce to the graph heat equation:

$$\dot{\boldsymbol{\theta}} = -K\mathbf{L}\boldsymbol{\theta} \implies \boldsymbol{\theta}(t) = e^{-K\mathbf{L}t}\boldsymbol{\theta}(0). \quad (\text{A.1})$$

Assuming independent Gaussian initial conditions $\boldsymbol{\theta}(0) \sim \mathcal{N}(\mathbf{0}, \sigma^2\mathbb{I})$, the mean phase vector is $\langle \boldsymbol{\theta}(0) \rangle = \mathbf{0}$ and the covariance is $\langle \boldsymbol{\theta}(0)\boldsymbol{\theta}(0)^\top \rangle = \sigma^2\mathbb{I}$. Because $\boldsymbol{\theta}(t)$ is a linear transformation of $\boldsymbol{\theta}(0)$, $\boldsymbol{\theta}(t)$ remains a zero-mean Gaussian random vector with covariance matrix:

$$\begin{aligned} \mathbf{C}(t) &\equiv \langle \boldsymbol{\theta}(t)\boldsymbol{\theta}(t)^\top \rangle = e^{-K\mathbf{L}t} \langle \boldsymbol{\theta}(0)\boldsymbol{\theta}(0)^\top \rangle e^{-K\mathbf{L}^\top t} \\ &= \sigma^2 e^{-2K\mathbf{L}t} = \sigma^2 \sum_{\alpha=1}^N e^{-2K\lambda_\alpha t} \mathbf{v}_\alpha \mathbf{v}_\alpha^\top. \end{aligned} \quad (\text{A.2})$$

Phase Difference Distribution: For any pair of nodes (i, j) , the phase difference $\Delta_{ij}(t) = \theta_i(t) - \theta_j(t) = (\mathbf{e}_i - \mathbf{e}_j)^\top \boldsymbol{\theta}(t)$ is a linear projection of a Gaussian vector, and thus $\Delta_{ij}(t) \sim \mathcal{N}(0, s_{ij}^2(t))$ with variance:

$$s_{ij}^2(t) = (\mathbf{e}_i - \mathbf{e}_j)^\top \mathbf{C}(t) (\mathbf{e}_i - \mathbf{e}_j) = C_{ii}(t) + C_{jj}(t) - 2C_{ij}(t). \quad (\text{A.3})$$

Substituting the spectral decomposition:

$$s_{ij}^2(t) = \sigma^2 \sum_{\alpha=1}^N e^{-2K\lambda_\alpha t} (v_\alpha(i)^2 + v_\alpha(j)^2 - 2v_\alpha(i)v_\alpha(j)) = \sigma^2 \sum_{\alpha=1}^N e^{-2K\lambda_\alpha t} (v_\alpha(i) - v_\alpha(j))^2. \quad (\text{A.4})$$

Isolation of the Zero Eigenmode: For a connected graph, the smallest eigenvalue $\lambda_1 = 0$ is simple, with constant eigenvector $\mathbf{v}_1 = \frac{1}{\sqrt{N}}\mathbf{1}$. Therefore, for every pair (i, j) :

$$v_1(i) - v_1(j) = \frac{1}{\sqrt{N}} - \frac{1}{\sqrt{N}} = 0. \quad (\text{A.5})$$

The $\alpha = 1$ term identically vanishes for all t :

$$s_{ij}^2(t) = \sigma^2 \sum_{\alpha=2}^N e^{-2K\lambda_\alpha t} (v_\alpha(i) - v_\alpha(j))^2. \quad (\text{A.6})$$

Characteristic Function and Order Parameter: The pairwise order parameter $\rho_{ij}(t) = \langle \cos(\theta_i - \theta_j) \rangle$ is the real part of the characteristic function $\phi_{\Delta_{ij}}(1) = \langle e^{i\Delta_{ij}} \rangle$. For a zero-mean Gaussian variable with variance $s_{ij}^2(t)$:

$$\langle e^{i\Delta_{ij}} \rangle = \exp\left(-\frac{1}{2}s_{ij}^2(t)\right). \quad (\text{A.7})$$

Because this expression is strictly real:

$$\rho_{ij}(t) = \text{Re}\langle e^{i\Delta_{ij}} \rangle = \exp\left(-\frac{1}{2}s_{ij}^2(t)\right). \quad (\text{A.8})$$

Comparing with the definition of the graph diffusion distance $d_t^2(i, j) = \|\Psi_t(i) - \Psi_t(j)\|^2$ in the diffusion map $\Psi_t(i) = \sigma(e^{-K\lambda_2 t}v_2(i), \dots, e^{-K\lambda_N t}v_N(i))$ [8]:

$$d_t^2(i, j) = \sigma^2 \sum_{\alpha=2}^N e^{-2K\lambda_\alpha t} (v_\alpha(i) - v_\alpha(j))^2, \quad (\text{A.9})$$

we confirm $s_{ij}^2(t) = d_t^2(i, j)$, completing the proof:

$$\rho_{ij}(t) = \exp\left(-\frac{1}{2}d_t^2(i, j)\right). \quad \square \quad (\text{A.10})$$

A.2 | Block Structure of the Tangled Nature Community Jacobian

Partitioning the S populated genotypes into the autonomous mutualistic core (C , size $S_{\text{core}} = 4$) and the passive mutational halo (H , size $S_{\text{halo}} = 82$), the community Jacobian K partitions into four blocks:

$$K = \begin{pmatrix} K_{CC} & K_{CH} \\ K_{HC} & K_{HH} \end{pmatrix}. \quad (\text{A.11})$$

- (i) **Core-Core Block (K_{CC}):** Governed by strong mutualistic cross-couplings and carrying-capacity damping. Its four eigenvalues lie deep in the left half-plane ($|\lambda| \in [26, 105]$), reflecting rapid collective relaxation ($\tau < 0.04$ generations).

- (ii) **Halo-Halo Block** (K_{HH}): Because halo species have negligible abundances ($n_h \ll N$) and zero autonomous reproduction ($p_{\text{off}} \leq 10^{-32}$), the derivative $\partial \dot{n}_h / \partial n_{h'}$ vanishes for all $h \neq h'$, while the diagonal simplifies to demographic relaxation:

$$K_{hh} = 2p_{\text{off}}(H_h) - p_{\text{kill}} \approx -p_{\text{kill}} = -0.20 \quad (\text{or } -1.0 \text{ in generational units}). \quad (\text{A.12})$$

Hence $K_{HH} \approx -p_{\text{kill}} \mathbb{I}_{82}$ is degenerate.

- (iii) **Halo-Core Block** (K_{HC}): Halo abundances are driven by mutational leakage from core parents. Differentiating the mutation influx $n_h^{\text{in}} = \sum_c M(c \rightarrow h)p_{\text{off}}(H_c)n_c$ generates directed coupling from core to halo, but no reciprocal feedback from halo to core ($K_{CH} \approx \mathbf{0}$).

Consequently, the system exhibits a triangular block structure: perturbations to halo genotypes do not propagate to the core, explaining why the empirical response manifold is continuous and governed by single-node demographic decay rather than collective low-dimensional attractors.

B | Bibliography

- [1] A. Arenas, A. Díaz-Guilera, and C. J. Pérez-Vicente. Synchronization reveals topological scales in complex networks. *Phys. Rev. Lett.*, 96:114102, 2006. doi: 10.1103/PhysRevLett.96.114102.
- [2] A. Arenas, A. Díaz-Guilera, J. Kurths, Y. Moreno, and C. Zhou. Synchronization in complex networks. *Phys. Rep.*, 469:93–153, 2008. doi: 10.1016/j.physrep.2008.09.002.
- [3] Rudy Arthur. Tangled-nature simulator. <https://github.com/rudyarthur/Tangled-Nature>, 2020.
- [4] Giacomo Barzon, Oriol Artime, Samir Suweis, and Manlio De Domenico. Unraveling the mesoscale organization induced by network-driven processes. *Proceedings of the National Academy of Sciences*, 121(23):e2317608121, 2024.
- [5] Nikolaj Becker and Paolo Sibani. Evolution and non-equilibrium physics: A study of the Tangled Nature model. *EPL (Europhysics Letters)*, 105(1):18005, 2014. doi: 10.1209/0295-5075/105/18005.
- [6] Andrea Cairoli, Duccio Piovani, and Henrik Jeldtoft Jensen. Forecasting transitions in systems with high-dimensional stochastic complex dynamics: A linear stability analysis of the Tangled Nature model. *Physical Review Letters*, 113(26):264102, 2014. doi: 10.1103/PhysRevLett.113.264102.
- [7] Kim Christensen, Simone A Di Collobiano, Matthew Hall, and Henrik Jeldtoft Jensen. Tangled nature: a model of emergent structure and temporal mode among co-evolving agents. *Journal of Theoretical Biology*, 216(1):73–84, 2002.
- [8] R. R. Coifman and S. Lafon. Diffusion maps. *Appl. Comput. Harmon. Anal.*, 21(1):5–30, 2006. doi: 10.1016/j.acha.2006.04.006.
- [9] Leon Danon, Albert Diaz-Guilera, Jordi Duch, and Alex Arenas. Comparing community structure identification. *J. Stat. Mech.*, 2005:P09008, 2005.
- [10] Jianxi Gao, Baruch Barzel, and Albert-László Barabási. Universal resilience patterns in complex networks. *Nature*, 530(7590):307–312, 2016.
- [11] S. Maslov and K. Sneppen. Specificity and stability in topology of protein networks. *Science*, 296(5569):910–913, 2002. doi: 10.1126/science.1065103.

- [12] Priodyuti Pradhan, Alok Yadav, Sanjiv K. Dwivedi, and Sarika Jalan. Optimized evolution of networks for principal eigenvector localization. *Phys. Rev. E*, 96(2): 022312, 2017. doi: 10.1103/PhysRevE.96.022312.
- [13] R. Sibson. SLINK: an optimally efficient algorithm for the single-link cluster method. *Comput. J.*, 16(1):30–34, 1973. doi: 10.1093/comjnl/16.1.30.
- [14] Lloyd N Trefethen and Mark Embree. *Spectra and Pseudospectra: The Behavior of Nonnormal Matrices and Operators*. Princeton University Press, 2005.